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Impact of adult neurogenesis in the dentate gyrus on CA3 memory functions: a computational model of its temporal dynamics

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Abstract

Adult neurogenesis in the hippocampal dentate gyrus (DG) is implicated in diverse memory functions, ranging from encoding and consolidation to forgetting. Despite extensive experimental evidence on its macro-level effects, the computational mechanisms by which immature granule cells (iGCs) generated through adult neurogenesis modulate the hippocampal circuit remain poorly understood. Studies on the topic often focus on local DG effects, neglecting the impact on downstream regions such as Cornu Ammonis 3 (CA3), and treat iGCs as a static population, a simplification that fails to capture the dynamics of their maturation. This project proposes the development of a DG-CA3 computational model based on experimental data to investigate these questions. To this end, the Brian2 simulator will be employed, using Izhikevich neurons and synapses endowed with both short-term plasticity and spike-timing-dependent long-term plasticity. The study will analyze how iGCs, during their peak excitability phase and after their maturation into mature granule cells (mGCs), influence not only pattern separation in the DG, but also the autoassociation and pattern completion functions of CA3. The model will simulate the temporal maturation of iGCs and the continuous addition of new neurons, as well as incorporate CA3-to-DG feedback projections, thereby aiming for a deeper understanding of the mechanisms by which neurogenesis modulates hippocampal circuit dynamics. It is expected that the DG will exhibit a mixed coding scheme, with mGCs specializing in pattern separation and iGCs in integration. It is also hypothesized that the high excitability of iGCs will, on one hand, impair CA3 pattern completion via non-selective assembly activation and, on the other hand, enable the temporal integration of information across their maturation, thus modulating memory consolidation.

Keywords: Adult neurogenesis, hippocampus, computational model, plasticity, dentate gyrus, CA3.

Resumo

A neurogênese adulta no giro denteado (DG) do hipocampo está implicada em diversas funções da memória, desde a codificação e consolidação até o esquecimento. Apesar das evidências experimentais sobre seus efeitos macro, os mecanismos computacionais pelos quais as células granulares imaturas (iGCs) geradas pela neurogênese adulta modulam o circuito hipocampal permanecem pouco compreendidos. Estudos sobre o tema frequentemente se concentram nos efeitos locais no DG, negligenciando o impacto em áreas subsequentes como o Cornu Ammonis 3 (CA3), e tratam as iGCs como uma população estática, uma simplificação que não captura a dinâmica de sua maturação. Este projeto propõe o desenvolvimento de um modelo computacional do circuito DG-CA3 baseado em dados experimentais para investigar essas questões. Para isso, será utilizado o simulador Brian2, com neurônios Izhikevich, sinapses com plasticidade de curto prazo e de longo prazo dependente do tempo de disparo. O trabalho analisará como as iGCs, durante sua fase de maior excitabilidade e após sua maturação em células maduras (mGCs), influenciam não apenas a separação de padrões do DG, mas também as funções de autoassociação e completamento de padrões do CA3. O modelo simulará a maturação temporal das iGCs e a adição contínua de novos neurônios, além de incorporar as retroprojeções do CA3 para o DG, buscando assim uma compreensão mais aprofundada dos mecanismos pelos quais a neurogênese modula a dinâmica do circuito hipocampal. Espera-se que o DG apresente uma codificação mista, onde mGCs se especializam na separação de padrões e iGCs na integração. Acredita-se, também, que a alta excitabilidade das iGCs, por um lado, prejudique o completamento de padrões no CA3 através da ativação promíscua de assembleias e, por outro, possibilite a integração temporal de informações ao longo de sua maturação, modulando a consolidação de memórias.

Palavras-chave: Neurogênese adulta, hipocampo, modelo computacional, plasticidade, giro denteado, CA3.

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Sumário

1 Introduction

To the hippocampus is attributed a fundamental role in the encoding, consolidation, and recall of declarative memories (Eichenbaum, 1999; Liao & Losonczy, 2024). To this end, the hippocampus receives afferent inputs from multiple cortical association areas, where information from diverse sensory modalities is processed. This information converges through the parahippocampal region, which comprises the perirhinal, postrhinal, and, most importantly, the entorhinal cortex (EC) (Eichenbaum, 2000). This rich convergence of information positions the hippocampus ideally to encode associations between events, consolidating its central role in learning and memory (Berdugo-Vega et al., 2023; Henke, 2010).

In a simplified and traditional account, the structure of the hippocampus has been described as consisting primarily of the trisynaptic circuit: information flows in a single direction from the EC to the dentate gyrus (DG), to CA3, and to CA1. The EC serves as the main input to the DG via the perforant path (PP), whose neurons connect primarily with Granule Cells (GCs), the principal excitatory cells of the DG, in a ratio of $1:10$; it is worth noting that the EC also sends afferent inputs to CA3 and CA1, which challenges the simplified notion of the trisynaptic circuit (Basu & Siegelbaum, 2015). Precisely because of this divergent EC-to-DG connectivity, it is believed that the DG is responsible for a computational function called pattern separation, through its sparse activity and the spike rate of GCs (Hainmueller & Bartos, 2020; Kesner & Hopkins, 2006; Yassa & Stark, 2011). Pattern separation disambiguates similar information, enabling the identification and differentiation of distinct yet highly similar experiences, which is fundamental for avoiding catastrophic interference between memories stored in CA3 (Rolls, 2013).

To illustrate this computation, consider the following oversimplified example: experiences from the same place, with the same people and objects present and following a similar routine, are likely encoded in the EC in a very similar manner; however, owing to pattern separation performed by the DG, small details such as a specific conversation or a news item on television may be sufficient to elicit a neural activity pattern in the DG that is very distinct from those of past experiences, allowing the encoding of a new memory (Eichenbaum, 2004).

In contrast to the divergent projection from the EC to the DG, the projection from the DG to CA3 is highly convergent. The CA3 region is characterized by a network of recurrent connections, where its pyramidal CA3 cells (pCA3) excite both interneurons and other pCA3 cells. This architecture forms an autoassociative network that allows the storage of patterns as neural assemblies: groups of neurons that represent the same pattern by spiking together, formed through Spike-Timing-Dependent Plasticity (STDP), which potentiates the connections of neurons that spike together (Kopsick et al., 2024). Once an assembly is formed through the strengthening of synapses between its cells, it can subsequently be reactivated by a partial input signal from the original pattern, since activating a subpopulation of assembly cells will cause the remaining cells to become active due to the associations formed by synaptic potentiation. This process is known as pattern completion, and recent computational models demonstrate that this architecture allows robust memory retrieval even from degraded cues (Kopsick et al., 2024; Le Duigou et al., 2014). Via the Schaffer collaterals, CA3 projects to CA1, which acts as an interface between the hippocampus and the neocortex, where memories are consolidated and recalled (Bartsch et al., 2011).

Beyond its fundamental role in pattern separation, the DG is one of the only areas of the mammalian brain to exhibit adult neurogenesis, the formation of new neurons from stem cells following development, continuously generating Granule Cells throughout life (Boldrini et al., 2018; Dumitru et al., 2025). Adult neurogenesis in mammals occurs only in a few other specific areas, such as the olfactory bulb in rodents and possibly the striatum, amygdala, and hypothalamus in humans (Alonso et al., 2024; Jurkowski et al., 2020).

The GCs generated by adult neurogenesis take several weeks to fully develop in rodents. At 3 weeks, the vast majority of immature Granule Cells (iGCs) die, and from approximately 8 weeks onward they become practically indistinguishable from mature Granule Cells (mGCs) (Denoth-Lippuner & Jessberger, 2021). Over the course of maturation, iGCs undergo various morphological and electrophysiological changes, expanding their dendrites and forming connections with CA3. A critical period occurs at approximately 4–6 weeks, during which iGCs, already partially integrated into the circuit, exhibit greater excitability and synaptic plasticity compared to mGCs (Aimone et al., 2014; Denoth-Lippuner & Jessberger, 2021; Zhao et al., 2006), offering a critical window for plasticity and memory encoding (Berdugo-Vega et al., 2023).

In general, adult neurogenesis is implicated in all phases of memory, including encoding, consolidation, recall, and even forgetting (Chavan et al., 2025). iGCs play a crucial role in novelty detection and pattern separation, enabling the discrimination between similar memories. Furthermore, recent evidence suggests a specialized role for these neurons in memory consolidation during REM sleep (Chavan et al., 2025) and in the consolidation process that generalizes memories, being essential for structural changes in CA3 circuits through the non-selective activation of neural assemblies via their high excitability (Ko et al., 2025).

Despite numerous studies demonstrating the role of adult neurogenesis in different hippocampal functions (Berdugo-Vega et al., 2023; Chavan et al., 2025), the computational mechanisms at the circuit level through which iGCs perform these functions remain poorly understood.

This central question unfolds into two main theoretical lines regarding the function of iGCs (Berdugo-Vega et al., 2023). The first hypothesis is that iGCs participate directly in memory encoding, integrating directly into long-term memory engrams and fulfilling a role similar to mGCs, but with their own peculiarities. In this scenario, they would act in the encoding of new information, in pattern integration, or in the prevention of catastrophic interference with memories already established by mGCs. The second hypothesis, which has been explored more recently, proposes that iGCs function as temporary circuit-modulating elements. In this role, they could both regulate DG activity, increasing the sparsity of activation by inhibiting mGCs, and modulate CA3 activity (Aimone, 2016; Berdugo-Vega et al., 2023), rather than being directly recruited in memory encoding as mGCs are.

Given the difficulty of experimentation in the DG and the complexity of information processed in the hippocampus, computational models have been invaluable for understanding the role of neurogenesis (Aimone, 2016). Computational models vary in scale, biological plausibility, the circuits modeled, and various other factors; therefore, a range of different results exists regarding the function and mechanisms of neurogenesis (Aimone, 2016; Berdugo-Vega et al., 2023).

In a conductance-based DG model (Kim & Lim, 2024b), the large population of mGCs performed excellent pattern separation, while the minority of iGCs served as pattern integrators. The presence of iGCs deteriorated the DG's pattern separation capacity compared to a DG composed solely of mGCs; however, the authors postulate that this mixed coding of pattern integration and separation could increase the capacity for memory storage and retrieval, a hypothesis they were unable to test since only the DG was modeled, without CA1 or CA3. Interestingly, these results contradict experimental studies demonstrating that an increase in neurogenesis enhances pattern separation (Sahay et al., 2011), which may indicate that computational models of the hippocampus have not yet fully captured the computations it performs.

In contrast, the work of Yang et al. (2025) employed a DG model to investigate the dynamic impact of neurogenesis on pattern separation. Their results indicate that the general effect of iGCs on GCs is inhibitory, which increases network sparsity and consequently improves pattern separation, providing computational evidence for the "indirect coding" hypothesis. The authors also investigated how iGC characteristics during maturation, such as decreasing activity and increasing synaptic plasticity, dynamically affect pattern separation. The study further suggests that the presence of iGCs increases the adaptability of the DG network to different input frequencies, improving the robustness of pattern separation.

The model of Aimone et al. (2009) was implemented at an intermediate level of biological plausibility[1], which allowed for the investigation of iGCs throughout their entire maturation over time. By modeling the maturation timeline of this iGC population, the model showed that iGCs preferentially encoded information received when they were young, thus exhibiting the role of temporal integrators. This model was corroborated by experimental studies (Berdugo-Vega et al., 2023), but did not analyze the implications for other areas such as CA1 and CA3, modeling only

[1] This model is considered to be at an intermediate level of biological plausibility because it does not model the electrical dynamics of individual neurons through conductance-based models, as in this project, but instead uses a digitized spike-rate model with a time step of 25 ms.

the DG.

This master's project proposes the development of a computational model of the DG-CA3 circuit to investigate the function of adult neurogenesis. Unlike many works that focus solely on the DG or on static populations of immature cells, this study will adopt a more holistic and thus innovative approach. The model will simulate the temporal maturation of iGCs into mGCs, enabling a dynamic analysis of how the continuous integration of new neurons affects the circuit over time. The main objective is to analyze the impacts of neurogenesis not only on DG pattern separation, but also on the autoassociation and pattern completion functions of the CA3 region. The effects of different levels of iGC integration, their heightened excitability, and how CA3-to-DG feedback projections, a feedback mechanism frequently neglected in models (Myers & Scharfman, 2011), influence network dynamics will be investigated. In doing so, the aim is to understand how iGCs, during their immature phase and after maturation, modulate the formation and reactivation of memories in CA3.

2 Justification

Although numerous works have attempted to define a function for neurogenesis, the role played by the immature cells it generates within the circuit they integrate into, as well as what occurs to that circuit during their maturation, remains poorly understood (Aimone, 2016; Aimone & Gage, 2011; Berdugo-Vega et al., 2023; Fares et al., 2019).

A number of studies on this topic have focused primarily on the effects of neurogenesis in the DG itself and on the pattern separation it performs (Berdugo-Vega et al., 2023; Kim & Lim, 2024a; Wang et al., 2024), neglecting its impact on connected regions such as the CA3 area. This project therefore aims to fill this gap by also examining how iGCs affect the memory functions of the CA3 area (autoassociation and pattern completion), and how the feedback projections from this area to the DG influence the system (Myers & Scharfman, 2011), with the goal of producing a more holistic and biologically plausible model.

It is also common for a number of works to study the function of a static iGC population (Aimone, 2016; Berdugo-Vega et al., 2023), with models that account for the maturation of these cells over time and the replacement of mGCs being very rare (Aimone et al., 2009). Such studies can explain the function (or lack thereof) of these immature cells only while they remain immature, but are unable to explain how this phase impacts these cells as they mature.

This study is therefore justified by its innovative approach, filling gaps in the existing literature by simulating the maturation of iGCs and its consequences in CA3.

3 Objectives

3.1 General objective

To analyze the impacts of adult neurogenesis on the capacity for autoassociation and pattern completion and separation in the DG-CA3 circuit in a computational model based on experimental data.

3.2 Specific objectives

- To analyze the effects of adult neurogenesis on DG pattern separation under different conditions of input pattern similarity.
- To consider the effects of incomplete iGC circuit integration, simulating different levels of connectivity.
- To analyze the effects of iGC circuit integration on CA3 autoassociation and pattern completion, as well as their role in neural assembly activation.
- To simulate the maturation of GCs generated by adult neurogenesis over time.

4 Hypotheses

Based on preliminary results and the literature, the central hypothesis of this work is that the computational model will reveal a functional dichotomy between Granule Cell populations. It is expected that mGCs will behave as pattern separators, increasing the dissimilarity between the neural representations of distinct stimuli. In contrast, iGCs, by virtue of their greater intrinsic excitability, are predicted to function as pattern integrators, with their non-selective activation decreasing the distinction between output patterns and, consequently, degrading the overall pattern separation capacity of the DG.

Additionally, it is hypothesized that this indiscriminate activation of iGCs will negatively impact CA3 memory functions, impairing the fidelity of pattern completion by co-activating neural assemblies unrelated to the input stimulus. Although iGC activity may suppress that of mGCs through shared inhibitory circuits, it is expected that this suppression will marginally increase mGC sparsity, slightly improving their separation capacity. Finally, the model will test the hypothesis that iGCs act as temporal integrators, with their maturation specializing them to respond to information encoded during their immature phase, establishing a mechanism for the separation of memories in time.

5 Materials and Methods

5.1 DG-CA3 neural network model

Based primarily on the models of Chavlis et al. (2017), Kim e Lim (2024a), Kopsick et al. (2024) e Yang et al. (2025), the DG-CA3 network architecture was modeled as shown in Figures 1 and 2 at a scale of $\frac{1}{500}$ of the rat hippocampus, with the number of neurons per population described in Table 1. Neurons were modeled simply, with a single compartment, but with high biological plausibility, using neuron and synapse models and cell count and connectivity data from Hippocampome.org, an open-access knowledge base that compiles multiple data sources on the hippocampal formation (Wheeler et al., 2023).

The network input is composed of cells from the entorhinal cortex (EC), with a total of $N_{EC} = 400$ neurons (Amaral et al., 1990; Kim & Lim, 2024a). In each simulation, the EC as a whole presents a specific pattern, where each pattern is represented by an active subpopulation of 10% of EC neurons (McNaughton et al., 1991). Inactive neurons not belonging to the pattern do not spike during the simulation, whereas active neurons spike according to a Poisson distribution with a spike rate of $\lambda = 40\,\text{Hz}$. EC neurons project their axons via the perforant path to neurons with dendrites in the molecular layer of the DG: Granule Cells (GCs), Mossy Cells (MCs) (Scharfman & Myers, 2013), and Basket Cells (BCs); as well as to CA3 neurons.

GCs are subdivided into two subpopulations: mature Granule Cells (mGCs) and immature Granule Cells (iGCs), representing GCs generated by adult neurogenesis at their characteristic electrophysiological phase of 4–6 weeks of age (Aimone et al., 2014). In total, the network is composed of $N_{GC} = 2000$ GCs ($\frac{1}{500}$ of the 10^6 rat granule cells) (West et al., 1991), with 5% of them being iGCs (Cameron & Mckay, 2001), giving $N_{mGC} = 1900$ and $N_{iGC} = 100$. Following the lamellar organization of the DG (Sloviter & Lømo, 2012), GCs are distributed across 20 lamellae, with 100 cells per lamella. Each GC connects with the BCs, MCs, and CA3 neurons of the same lamella, and also forms random, non-lamellar excitatory connections with Hilar Perforant Path-associated cells (HIPP).

iGCs, unlike mGCs, possess distinct electrophysiological characteristics that render them more excitable than mGCs. Furthermore, to simulate their not-yet-complete circuit integration, iGCs receive EC afferent inputs at only a fraction $x_{EC:iGC}$ of the connection probability between the EC and mGCs. Hippocampome.org (Wheeler et al., 2023) does not provide iGC characteristics. Therefore, the parameters used for iGCs were fitted from the spiking data of Espósito et al. (2005) using the method of Nelder e Mead (1965) (Table 2).

BCs ($N_{BC} = 40$) are inhibitory cells that ensure sparse GC activation through winner-take-all competition among them (Chavlis et al., 2017; Coultrip et al., 1992; Kim & Lim, 2024a), where only the most active GCs in a lamella remain active, inhibiting the rest. HIPPs

Figura 1 – DG-CA3 network architecture. Inhibitory synapses are represented by circles and excitatory ones by arrows. The network input is composed of cells from the entorhinal cortex (EC). The dentate gyrus (DG) has the perforant path (PP) and the dendrites of its cells in the molecular layer; the granular layer is composed of immature and mature Granule Cells (iGCs and mGCs, respectively) and Basket Cells (BCs); the hilus is composed of Mossy Cells (MCs) and Hilar Perforant Path-associated cells (HIPP). CA3 is composed of pyramidal CA3 cells (pCA3) and inhibitory CA3 cells (iCA3).

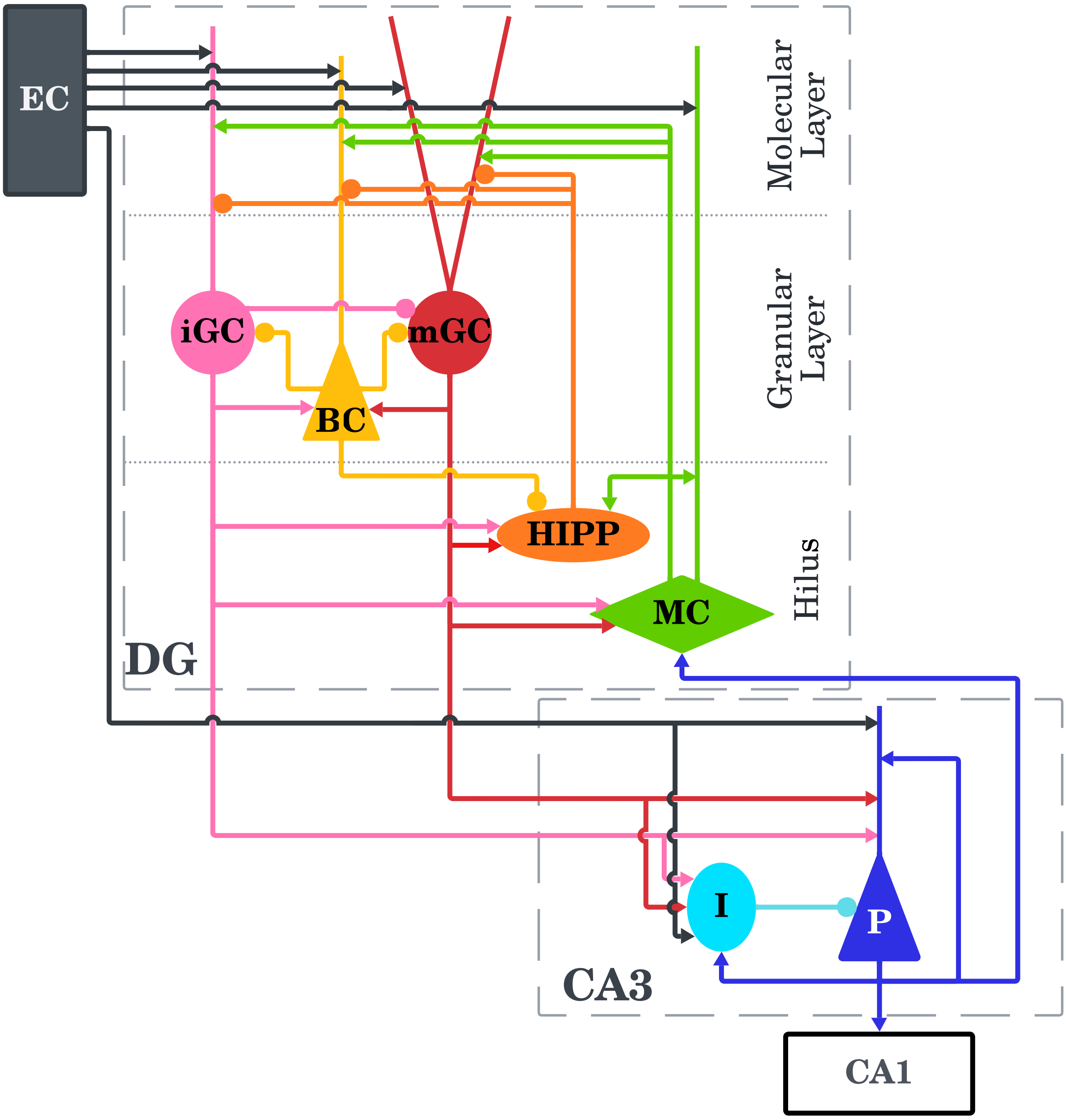
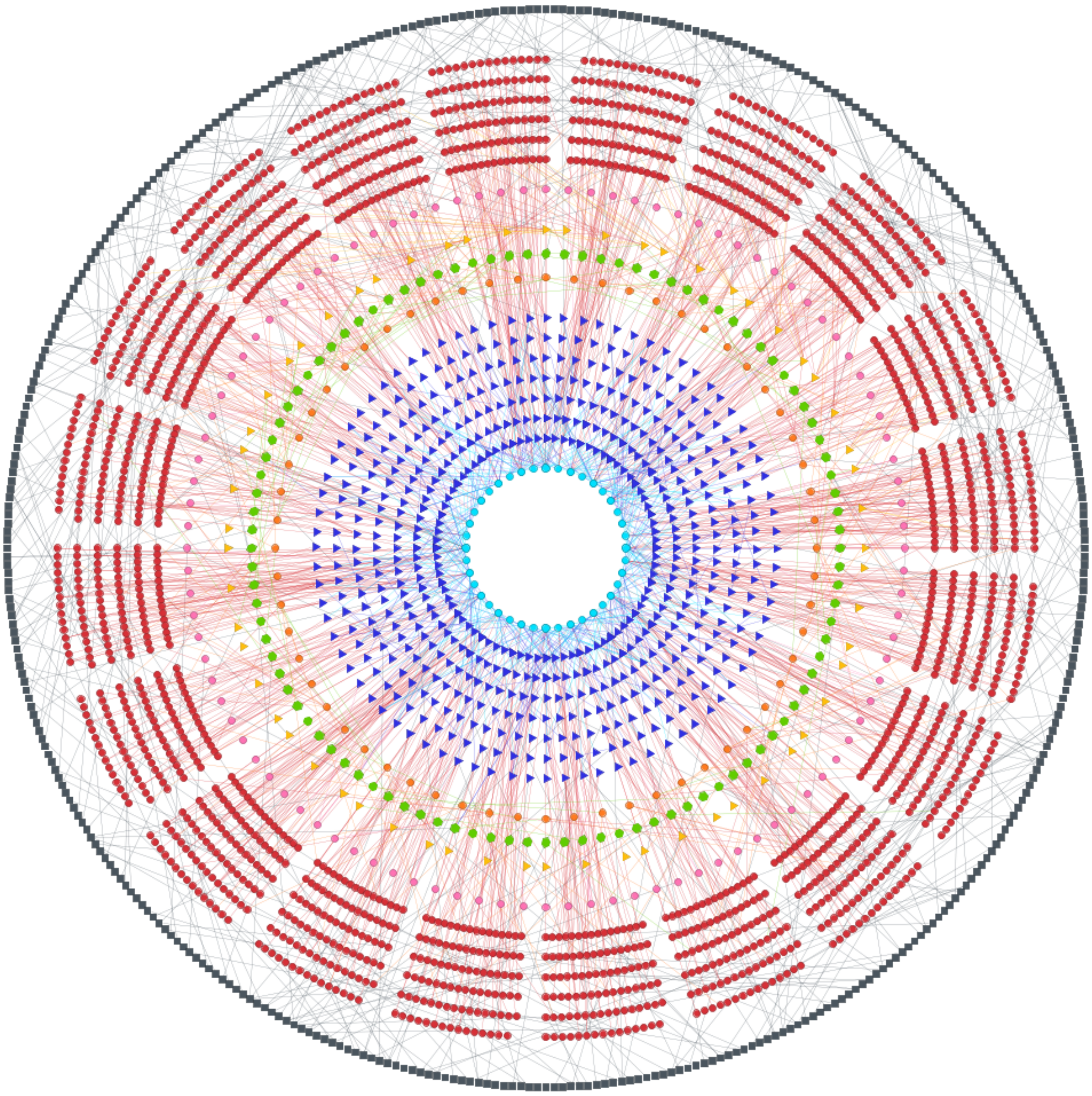


Figura 2 – Ring graph of the DG-CA3 network. Each node represents a neuron in the model, with the same color scheme as Figure 1 for each neuron. Each edge represents a synapse, colored according to the presynaptic neuron. The EC is represented by the outermost black circle; the lamellar organization of the DG can be seen with mGCs and iGCs (red and pink, respectively) and pCA3 cells at the center in teal. In total, the graph contains 3,260 nodes and 266,596 edges.



(N_{HIPP} = 60) also contribute to the sparsity of GC activation, with a more global inhibition acting on all GCs in a lamella. MCs (N_{MC} = 100) are excitatory hilar cells that receive excitation from GCs and project back to GCs, BCs, and HIPPs via interlamellar connections. Although these excitatory MC-to-GC projections exist, their net effect is generally inhibitory through the control of BCs and HIPPs (Myers & Scharfman, 2009; Scharfman & Myers, 2013).

The HIPP parameters in Hippocampome.org (Wheeler et al., 2023) model the "rebound burst spiking" characteristic, in which the neuron produces a burst of spikes following the end of a hyperpolarization period. This model is incompatible with the network, as large hyperpolarizations lead to infinite spike bursts. Therefore, the parameters used for HIPPs were taken from Modak e Chakravarthy (2018).

CA3, unlike the DG, does not follow a lamellar structure in the model, as it forms a far more integrative neural network (Pak et al., 2022; Watson et al., 2025). The CA3 model comprises 600 pyramidal cells (N_{pCA3} = 600) and 60 inhibitory cells (N_{iCA3} = 60), modeled according to the physiological data for CA3 basket cells (Wheeler et al., 2023) so as to simplify the variety of inhibitory neurons present in CA3 (Kopsick et al., 2024). Both neuronal populations receive afferent inputs from the EC and from GCs, with iCA3 cells inhibiting pCA3 cells. pCA3 cells excite iCA3 cells and, importantly, form a recurrent network by connecting with other pCA3 cells, a fundamental characteristic for the autoassociation and pattern completion functions of CA3 (Kopsick et al., 2024; Rolls, 2013). pCA3 cells also send feedback projections to the DG, connecting with MCs in the model, a process that contributes to pattern separation (Myers & Scharfman, 2011).

All simulations were performed with Brian2 (Stimberg et al., 2019), using the 4th-order Runge-Kutta method with a fixed time step of 0.1 ms (Butcher, 1996). Neuron and synapse parameters can be found in Tables 2 and 3, respectively. The duration of a single simulation for one input pattern is 500 ms for the network to reach a steady state, with the pattern recording phase occurring subsequently for 1000 ms. Synaptic connections are kept fixed across all simulations.

Cell	**N**
Entorhinal cortex	400
Mature granule	1900
Immature granule	100
Mossy	100
HIPP	60
Basket	60
CA3 Pyramidal	600
CA3 Inhibitory	60

Tabela 1 – Neuronal population size (N).

5.2 Neuron model

Neurons were modeled according to the 9-parameter Izhikevich neuron model (Izhikevich, 2006, cap. 8) with a single compartment, without considering dendrites or axons. This model was chosen in this work and by Wheeler et al. (2023) for being capable of capturing the dynamic behavior of neurons across a wide variety of conditions with biological plausibility, comparable to the Hodgkin-Huxley model (Hodgkin & Huxley, 1952), while presenting a simpler and computationally more efficient mathematical formulation. The spike traces of neurons can be seen in Figure 3. The Izhikevich neuron model is described by the following equations:

$$C_m \frac{dV_m}{dt} = k(V_m - V_r)(V_m - V_t) - u + I \tag{5.1}$$

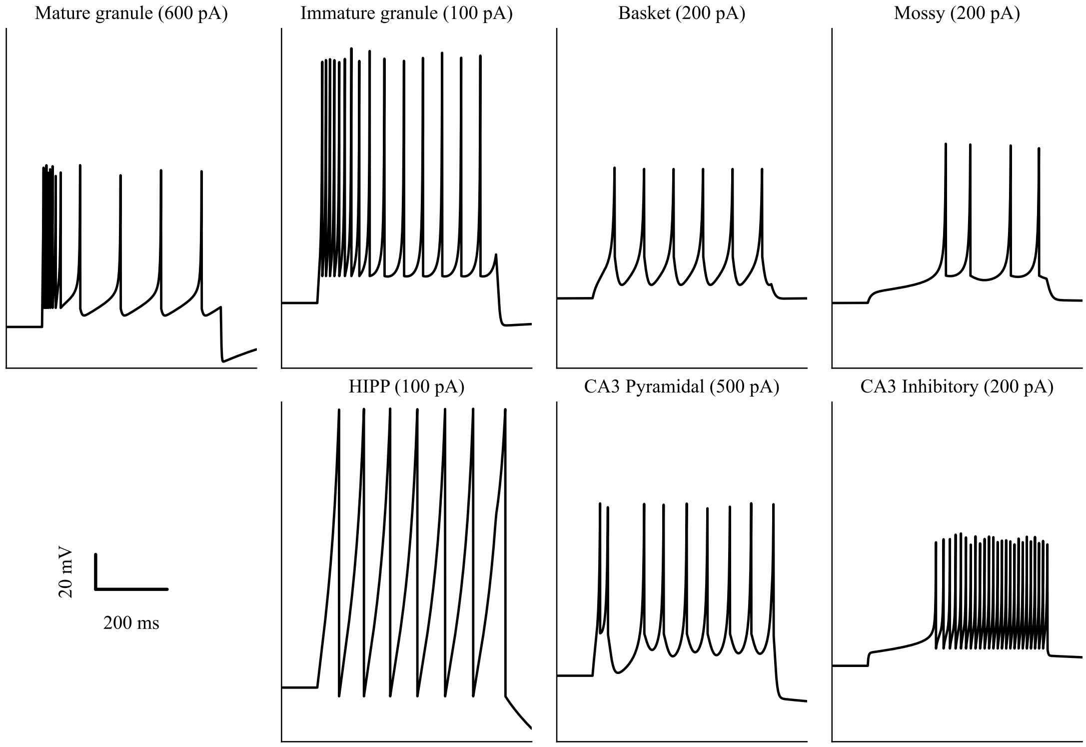
$$\frac{du}{dt} = a[b(V_m - V_r) - u] \tag{5.2}$$

Where V_m is the membrane potential, u is the recovery variable, C_m is the membrane capacitance, V_r is the resting potential, V_t is the threshold potential, I is the total current flowing into the neuron, and k, a, and b are constants that define the dynamic characteristics of the neuron. In addition to the differential equations above, which define the temporal evolution of the membrane potential and recovery variable, the Izhikevich neuron model also includes a rule for action potential generation, defined by Equation 5.3.

$$\text{if } V_m \geq V_{\text{peak}}, \quad \begin{cases} V_m \leftarrow V_{min} \\ u \leftarrow u + d \end{cases} \tag{5.3}$$

When the membrane potential reaches the peak value V_{peak}, an action potential is generated, the membrane potential is reset to the after-spike potential V_{min}, and the recovery variable u is incremented by d, making subsequent spike generation more difficult.

Figura 3 – Spike traces of neurons under constant current as indicated by each subfigure title.



Cell	k (nS/mV)	a (ms^{-1})	b (nS)	d (pA)	C_m (pF)	V_r (mV)	V_t (mV)	V_{min} (mV)	V_{peak} (mV)
Mature granule	0.45	0.003	24.48	50	38	-77.4	-44.9	-66.47	15.49
Immature granule	0.139	0.002	-1.877	12.149	24.6	-63.66	-38.41	-48.2	83.5
Mossy	1.5	0.004	-20.84	117	258	-63.67	-37.11	-47.98	28.29
HIPP	0.01	0.004	-2	40.52	58.7	-70	-50	-75	90
Basket	0.81	0.097	1.89	553	208	-61.02	-37.84	-36.23	14.08
CA3 Pyramidal	0.79	0.008	-42.55	588	186	-63.2	-33.6	-38.87	35.86
CA3 Inhibitory	1	0.004	-18.26	12	45	-57.51	-23.38	-47.56	18.45

Tabela 2 – Izhikevich model parameters per cell type.

5.3 Synapse model

The synapse model, like the neuron model, was defined based on Hippocampome.org (Wheeler et al., 2023), following the formulation of Senn et al. (2001)(Mongillo et al., 2008). This model captures short-term plasticity, whether short-term depression, caused by neurotransmitter depletion, or short-term potentiation, caused by calcium accumulation, both occurring on the scale of tenths of seconds. Each synapse has 5 parameters (described in Table 3): the maximum synaptic conductance in the absence of any synaptic resource depletion g, the proportion of resources utilized per spike U_{se}, the synaptic current decay time constant τ_d, the facilitation time constant τ_f, and the resource recovery time constant τ_r (Moradi et al., 2022). Another parameter shown in Table 3 is the connection probability P. Although Hippocampome.org provides data on connection probabilities between neuronal populations *in vivo*, the reduced scale of the network necessitated increasing these probabilities so that network activity could be maintained while sparse activity was preserved.

The model is described by three state variables: synaptic resource utilization (U), resource recovery (R), and the percentage of resources in an active state (A). Initially, $U_{t_0} = 0$, $R_{t_0} = 1$, and $A_{t_0} = 0$, since all resources are available for use. The temporal evolution of these variables is governed by the following system of differential equations:

$$\frac{dU}{dt} = \frac{-U}{\tau_f} + U_{se}(1 - U_-)\delta(\Delta t_i) \tag{5.4}$$

$$\frac{dR}{dt} = \frac{1 - R - A}{\tau_r} - U_+ R_- \delta(\Delta t_i) \tag{5.5}$$

$$\frac{dA}{dt} = \frac{-A}{\tau_d} + U_+ R_- \delta(\Delta t_i) \tag{5.6}$$

where δ is the Dirac delta function, yielding 1 only when $\Delta t_i = t - t_i = 0$, i.e., only at time t corresponding to the synaptic event time t_i. U_+ corresponds to the value of U immediately after the synaptic event, while R_- corresponds to the value of R immediately before the same event.

From these equations, the synaptic current is given by:

$$I = k \cdot A \cdot g \cdot (V_m - E) \tag{5.7}$$

where V_m is the postsynaptic membrane potential, E is the synaptic reversal potential, for inhibitory and excitatory synapses, respectively, $E_{inh} = -86\,\mathrm{mV}$ and $E_{exc} = 0\,\mathrm{mV}$, and k is a scaling constant defined as $k = 10$ for all synapses. This scaling constant is necessary due to the reduced scale of the network, since, given the low number of synapses in the model compared to the rat hippocampus, the entire network would fall silent without this scaling. All synaptic currents I of a neuron are summed and affect the membrane potential V_m according to Equation 5.1.

5.4 Long-term plasticity

Long-term plasticity will be implemented exclusively in the recurrent synapses between pyramidal CA3 cells (pCA3), given their fundamental role in the formation of neural assemblies and to ensure the computational efficiency of the model. The plasticity model will be based on the work of Kopsick et al. (2024), which employs a Spike-Timing-Dependent Plasticity (STDP) rule.

Presynaptic	**Postsynaptic**	**Connection**	P (%)	g (nS)	τ_d (ms)	τ_r (ms)	τ_f (ms)	U
Entorhinal cortex	Mature granule	Random	8	1.655	5.333	266.239	18.714	0.27
Entorhinal cortex	Immature granule	Random	4	1.655	5.333	266.239	18.714	0.27
Entorhinal cortex	Mossy	Random	20	1.522	4.671	319.835	57.766	0.204
Entorhinal cortex	Basket	Random	25	1.406	3.849	144.415	48.2	0.214
Entorhinal cortex	CA3 Pyramidal	Random	12	0.87	6.55	258.318	53.478	0.184
Entorhinal cortex	CA3 Inhibitory	Random	12	1.2	3.602	257.468	35.904	0.21
Mature granule	Mossy	Random	40	2.713	5.347	98.583	27.479	0.151
Mature granule	HIPP	Random	10	1.805	5.181	362.814	48.986	0.15
Mature granule	Basket	Lamellar	100	4.258	5.566	31.265	4.278	0.497
Mature granule	CA3 Pyramidal	Lamellar	80	1.28	6.657	278.286	78.584	0.155
Mature granule	CA3 Inhibitory	Lamellar	50	3.624	3.915	218.934	43.274	0.176
Immature granule	Mature granule	Random	5	0.524	3.915	218.934	43.274	0.176
Immature granule	Mossy	Random	40	2.713	5.347	98.583	27.479	0.151
Immature granule	HIPP	Random	10	1.805	5.181	362.814	48.986	0.15
Immature granule	Basket	Lamellar	100	4.258	5.566	31.265	4.278	0.497
Immature granule	CA3 Pyramidal	Lamellar	80	1.28	6.657	278.286	78.584	0.155
Immature granule	CA3 Inhibitory	Lamellar	50	4.224	3.915	218.934	43.274	0.176
Mossy	Mature granule	Cross-lamellar	0.2	1.894	5.357	166.162	20.224	0.304
Mossy	Immature granule	Cross-lamellar	0.2	1.894	5.357	166.162	20.224	0.304
Mossy	Mossy	Cross-lamellar	5	0.068	4.257	149.329	71.642	0.245
Mossy	HIPP	Cross-lamellar	50	2.376	4.824	358.431	54.872	0.181
Mossy	Basket	Cross-lamellar	100	1.996	3.396	117.365	69.316	0.255
HIPP	Mature granule	Random	30	3.202	8.935	159.143	10.396	0.278
HIPP	Immature granule	Random	10	3.202	8.935	159.143	10.396	0.278
HIPP	Basket	Random	2	1.709	5.982	367.198	15.292	0.221
Basket	Mature granule	Lamellar	100	4.351	3.543	103.876	12.347	0.332
Basket	Immature granule	Lamellar	10	4.351	3.543	103.876	12.347	0.332
Basket	HIPP	Random	2	1.408	6.544	534.182	8.385	0.24
CA3 Pyramidal	CA3 Pyramidal	Random	2.5	0.9	9.516	278.258	27.513	0.172
CA3 Pyramidal	Mossy	Lamellar	10	2.035	4.297	359.116	40.457	0.236
CA3 Pyramidal	CA3 Inhibitory	Random	20	2	4.525	275.604	23.321	0.189
CA3 Inhibitory	CA3 Pyramidal	Random	30	5	7.793	416.282	20.63	0.203

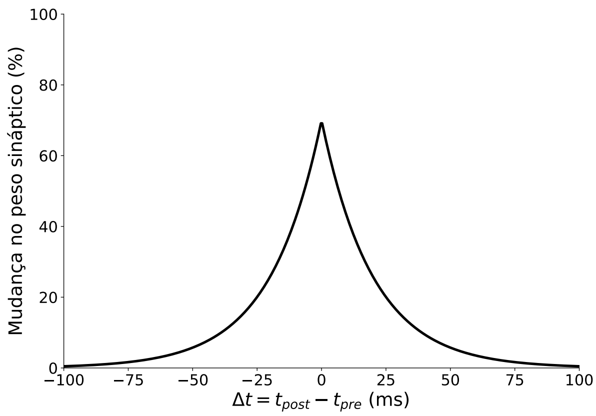
Tabela 3 – Parameters of the synapses between neuronal populations. Random connections occur between all cells of both populations; lamellar connections occur between cells of the same lamella; cross-lamellar connections occur between cells of one lamella and all others. The connection probability P refers to the percentage of connections between neuronal populations according to the connection condition.

Specifically, a symmetric, Hebbian STDP rule will be used, which promotes the strengthening of synapses between neurons that spike in close temporal proximity, regardless of spiking order. This rule is described by the following equation:

$$\Delta w = Ae^{-|\Delta t|/\tau} \tag{5.8}$$

Where Δw is the change in synaptic weight, A is a parameter that determines the maximum magnitude of the weight change, τ is the plasticity decay time constant, and Δt is the temporal difference between the spikes of the pre- and postsynaptic neuron. Following the model of (Kopsick et al., 2024), the time constant τ will be set to 20 ms. The parameter A will be adjusted to modulate the learning rate of the network.

Figura 4 – Symmetric Hebbian STDP rule with $A = 70$.



To prevent synaptic saturation, a synaptic weight renormalization rule (Section 5.7.1) and a maximum synaptic weight w_{max} to be determined experimentally will be employed.

5.5 Temporal neurogenesis

To investigate the functional impact of the continuous integration of new neurons, a temporal neurogenesis model will be implemented. This process is fundamental for the maintenance of hippocampal functions such as learning and memory over time (Aimone et al., 2014; Berdugo-Vega et al., 2023). The simulation will be divided into two main phases. In the first phase, the network will operate with its initial configuration, with $N_{mGC} = 1900$ and $N_{iGC} = 100$. During this phase, the network will be exposed to a set of N input patterns, which will be learned and stored in the recurrent synapses of CA3 via the STDP mechanism. The number of input patterns N will be varied experimentally to determine how many patterns the model can store without a significant deficit in pattern completion (see Section 5.7.2).

After the initial learning phase, iGCs will undergo a discrete simulated maturation process, transforming into mGCs all at once. This transition reflects the biological changes that occur as new neurons become fully integrated into the DG circuit, transitioning from a hyperexcitable state to a more stable mature state (Abbott & Nigussie, 2020). Maturation will be implemented by updating their electrophysiological and synaptic parameters to the values corresponding to those of mGCs (Tables 2 and 3). Existing efferent connections, formed during the immature phase, will be preserved. Additionally, the newly matured cells will have their EC afferent connectivity increased to the same level as mGCs, completing their functional integration into the circuit.

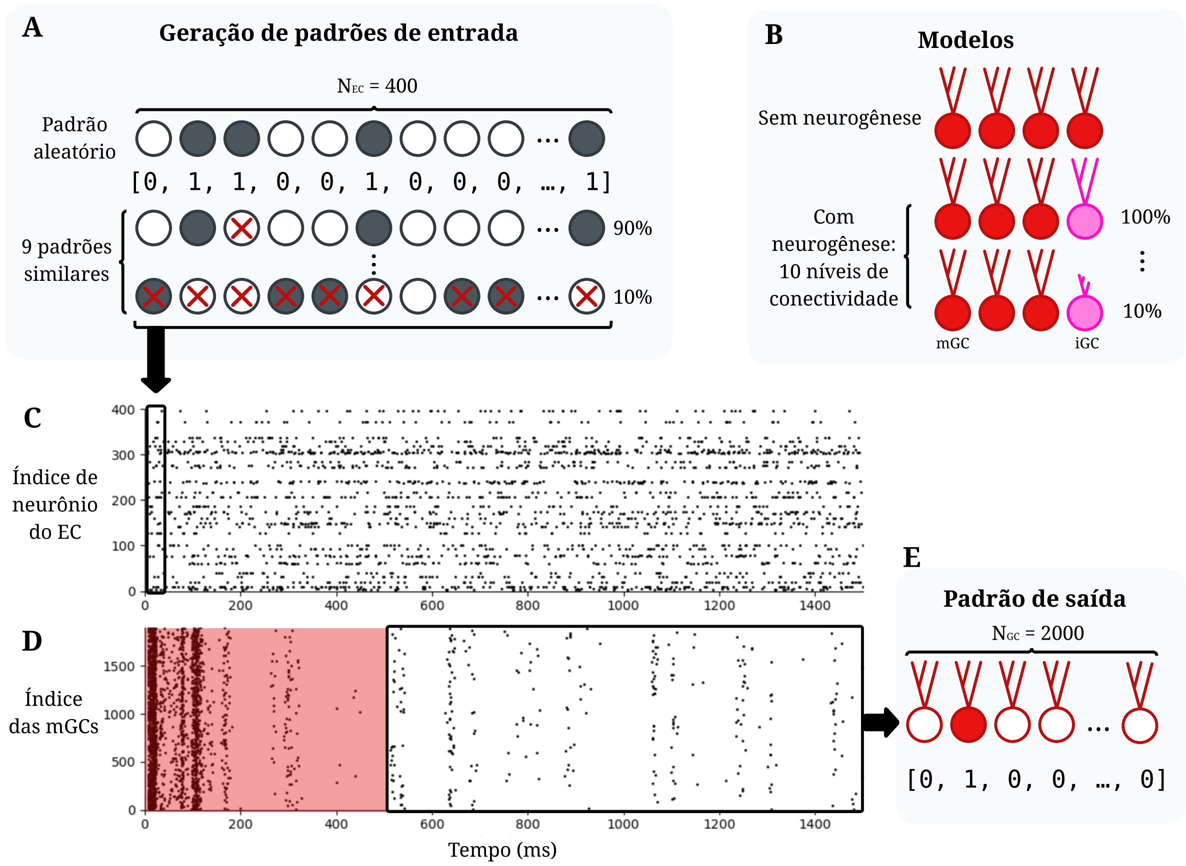
In the second phase of the simulation, following the maturation of the first iGCs, 100 new iGCs will be added, simulating the continuous integration of new neurons by adult neurogenesis, and another batch of N patterns will be presented to the network. This process will be repeated 3 times, making it possible to evaluate network performance progressively, tracking the maturation of iGCs. Two control models will be implemented: one in which only mGCs are added to the network without iGCs, and another in which, when iGCs are added, an equivalent number of mGCs undergo neuronal death. The purpose of these control models is to isolate the role of adult neurogenesis and of the immature cells it generates, comparing against the effects of control models that also feature a gradual increase in neuron count.

5.6 Pattern separation

The methodology for quantifying pattern separation was based on that used by Kim e Lim (2024b). To analyze pattern separation under different levels of input pattern similarity and the impact of adult neurogenesis, 11 different model variants were tested: one control without adult neurogenesis ($N_{mGC} = 2000$) and 10 others with neurogenesis at different connectivity levels ($N_{mGC} = 1900$ and $N_{iGC} = 100$), as illustrated in Figure 5. The connectivity level is expressed as a percentage, where 100% corresponds to iGCs having the same EC-GC connection probability as mGCs; the intent is to simulate the incomplete circuit integration of iGCs, whose dendrites are not yet fully developed. For each model, 20 sets of 10 input patterns are generated. Each set consists of one randomly generated original pattern and 9 patterns similar to it, ranging from 90% to 10% similarity. An input pattern consists of a binary vector of size $N_{EC} = 400$, representing the activation state of each EC neuron, in which active neurons spike according to a Poisson distribution at a frequency of 40 Hz. Every input pattern has exactly 10% active neurons. To generate patterns with different levels of similarity, a random fraction of X% of the active neurons is kept active, while the remaining $(100 - X)$% are deactivated and an equal number of initially inactive neurons is activated, to maintain the 10% activity level (40 active neurons).

Each pattern is simulated for 1500 ms of simulation time, with the first 500 ms discarded while the network stabilizes. The output pattern of a neuronal population, whether mGCs, iGCs, or pCA3 cells, consists of a binary vector with value 1 for the index of each neuron that spiked at least once during the 1000 ms interval.

Figura 5 – Diagram of the input and output pattern generation and recall protocol. **A.** Input patterns are generated in sets of 10 patterns of size $N_{EC} = 400$; the first is generated randomly and the remaining ones are generated from it at different levels of similarity by randomly activating and deactivating neurons. **B.** 11 model variants are simulated: one without neurogenesis and 10 with neurogenesis at different connectivity levels, from 100% down to 10% connectivity. **C.** Spike trace of EC neurons during a simulation; each generated pattern corresponds to the activation of EC neurons, with active neurons spiking at a frequency of 40 Hz; each of the 11 models is exposed to 20 sets of 10 input patterns, totaling 200 simulations of 1500 ms each. **D.** Spike trace of mGCs during a simulation; the first 500 ms are discarded while the network stabilizes; the remainder corresponds to the output pattern. **E.** The output pattern, illustrated here for the mGC population, consists of a binary vector with value 1 for the index of each neuron that spiked at least once during the 1000 ms interval.



To characterize pattern separation, the overlap between the neural activity patterns at the input (entorhinal cortex cells) and output (DG Granule Cells or CA3 pyramidal cells) of the network is compared. A pattern is defined by a binary representation of size N, where N is the total number of neurons in a specific population, with neurons that spiked at least once during the simulation time interval (1000 ms) represented by 1 and those that did not by 0. For a pair of patterns $A^{(l)}$ and $B^{(l)}$ (where $l \in \{in, out\}$ for input and output, respectively), the distance between patterns $D_p^{(l)}$ is defined as:

$$D_p^{(l)} = \frac{O^{(l)}}{D_a^{(l)}} \tag{5.9}$$

In this equation, $O^{(l)}$ represents the degree of orthogonalization and $D_a^{(l)}$ the mean degree of activation of the two patterns. The mean degree of activation $D_a^{(l)}$ is the arithmetic mean of the degrees of activation of each pattern, $A^{(l)}$ and $B^{(l)}$:

$$D_a^{(l)} = \frac{D_a^{(A^{(l)})} + D_a^{(B^{(l)})}}{2} \tag{5.10}$$

The degree of activation of an individual pattern is the fraction of active neurons (represented by 1 in a binary encoding) in the pattern. The orthogonalization degree $O^{(l)}$, which measures the dissimilarity between patterns, is calculated from the Pearson correlation coefficient, $\rho^{(l)}$:

$$O^{(l)} = \frac{1 - \rho^{(l)}}{2} \tag{5.11}$$

Where $\rho^{(l)}$ is the Pearson correlation coefficient between patterns $A^{(l)}$ and $B^{(l)}$. Considering $\{a_i^{(l)}\}$ and $\{b_i^{(l)}\}$ $(i = 1, \ldots, N_l)$ as the binary representations of the state of the i-th cell in patterns $A^{(l)}$ and $B^{(l)}$ $(l \in \{in, out\})$, the Pearson correlation coefficient is given by:

$$\rho^{(l)} = \frac{\sum_{i=1}^{N_l} \Delta a_i^{(l)} \cdot \Delta b_i^{(l)}}{\sqrt{\sum_{i=1}^{N_l} (\Delta a_i^{(l)})^2} \sqrt{\sum_{i=1}^{N_l} (\Delta b_i^{(l)})^2}} \tag{5.12}$$

where $\Delta a_i^{(l)} = a_i^{(l)} - \langle a^{(l)} \rangle$ and $\Delta b_i^{(l)} = b_i^{(l)} - \langle b^{(l)} \rangle$. The notation $\langle \ldots \rangle$ indicates the population mean over all cells. The value of $\rho^{(l)}$ varies between -1 and 1, therefore $O^{(l)}$ varies between 0 and 1 (Equation 5.11).

From the input ($D_p^{(in)}$) and output ($D_p^{(out)}$) pattern distances, the degree of pattern separation, $\mathcal{S}_D$, is calculated as the ratio between them:

$$\mathcal{S}_D = \frac{D_p^{(out)}}{D_p^{(in)}} \tag{5.13}$$

A value of $\mathcal{S}_D > 1$ indicates that the output patterns are more distinct than the input patterns, characterizing pattern separation. Conversely, $\mathcal{S}_D < 1$ indicates pattern convergence, where the output patterns become more similar to each other.

5.7 Autoassociation and pattern completion

5.7.1 Training and testing protocol

The network training and testing protocol for evaluating pattern storage and retrieval will be based on the method described by Kopsick et al. (2024), who implemented a full-scale CA3 model also using Hippocampome.org data and developed a biologically plausible training protocol.

During the training phase, the network will be exposed to a set of distinct input patterns. Each pattern consists of the activation of a specific subpopulation of pCA3 cells via current injection. This activation induces a spike train in a 20 ms time window, corresponding to one gamma cycle. The different patterns will be presented in sequence, separated by 200 ms windows, simulating a theta-gamma neural code. Throughout the training phase, synaptic plasticity (STDP) will be enabled, allowing the strengthening of connections between neurons encoding the same pattern and thus forming a neural assembly. After a number of pattern presentations to be determined experimentally, the synaptic weights between pyramidal cells will be renormalized, a process that simulates the synaptic homeostasis occurring during slow-wave sleep, to prevent synaptic saturation and stabilize the learned patterns (González-Rueda et al., 2018; Kopsick et al., 2024). This renormalization consists of dividing each synaptic weight between pyramidal cells by the same factor, such that the mean weight across all synapses returns to its initial value. This procedure preserves the relative weight distribution, thereby maintaining the learned associations without allowing synapses to saturate.

During the testing phase, synaptic plasticity will be disabled in order to assess the network's ability to retrieve the stored patterns without modification to the connections. Degraded versions of the original patterns will be presented, in which only a fraction of the neurons in the corresponding assembly is activated. The network response will then be analyzed as described in Section 5.7.2 to verify whether the activity of the recurrent connections can reconstruct the complete pattern through pattern completion.

5.7.2 Quantification

The strength of autoassociation, that is, the formation of assemblies, will be measured through two characteristics of the synapses between pyramidal cells. The first is the signal-to-noise ratio (SNR) of autoassociation, defined as the mean synaptic weight between neurons of the same assembly divided by the mean synaptic weight between neurons of different assemblies. A high SNR indicates a strong distinction between intra- and inter-assembly connections. The second metric is the percentage of synapses within an assembly that reached the maximum weight, which serves as an indicator of synaptic saturation.

Pattern completion will be assessed through a pattern reconstruction accuracy metric, based on (Kopsick et al., 2024). This metric uses the Pearson correlation coefficient (Equation 5.12) between input or output patterns during the training and testing phases.

First, the input pattern correlation, $\rho^{(in)}$, is calculated, which is the correlation between the complete EC pattern presented during the training phase and the degraded, or incomplete, pattern presented during the testing phase. Next, the output pattern correlation, $\rho^{(out)}$, is calculated, corresponding to the correlation between the pCA3 neural activity in response to the complete training pattern during assembly encoding and the activity in response to the degraded test pattern.

The pattern reconstruction accuracy, R_p, is then calculated as:

$$R_p = \frac{\rho^{(out)} - \rho^{(in)}}{1 - \rho^{(in)}} \tag{5.14}$$

A value of $R_p > 0$ indicates that pattern completion occurred, i.e., the output representation became more similar to the original pattern than the degraded input. A value of $R_p = 1$ represents perfect reconstruction.

6 Results

6.1 Pattern separation

Some initial results from the protocol described in Section 5.6 have already been obtained and will be discussed in this section.

Figure 6 presents the mean activation of GC populations for the control model (without neurogenesis) and for models with neurogenesis at different iGC-to-EC connectivity levels. In the control model, where all GCs are mature, the mean population activation, defined as the percentage of cells that spiked at least once during the 1000 ms simulation interval, was 12.46%, a level of sparse activity, though not as low as that typically observed in the DG. In future simulations, the impact of increasing inhibitory synapses in the circuit will be analyzed.

The introduction of iGCs into the circuit significantly alters this dynamics. As iGC connectivity increases, their activation rate rises sharply, consistent with their greater intrinsic excitability. At 100% connectivity, virtually the entire iGC population becomes active in response to an input pattern. This increase in iGC activity raises the overall GC population activation.

In contrast, a progressive decrease in mature Granule Cell (mGC) activity is observed as iGC connectivity increases. This result suggests the existence of an inhibitory competition mechanism, whereby iGCs, once strongly activated, indirectly suppress mGC activity through shared interneurons.

However, the suppression observed in mGCs is not as pronounced as that reported in other models (Kim & Lim, 2024b), which may again indicate that the inhibitory strength in the current model is insufficient. One hypothesis for this discrepancy is the absence of a direct inhibitory pathway from iGCs to mGCs, a mechanism described by Luna et al. (2019) that was not implemented. The inclusion of this direct connection is a promising avenue for future investigation, as it could confer greater biological plausibility to the model and will also be explored in the project.

Figura 6 – Mean population activation (%) per model. The vertical axis contains a break and two different scales for better visualization. Error is represented by the shaded area.

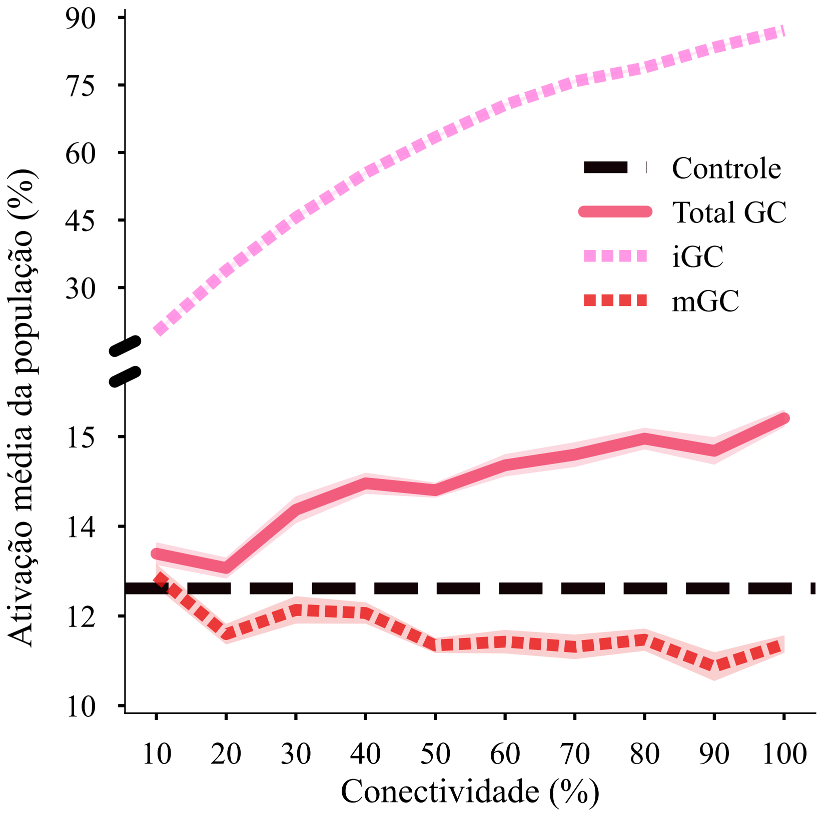
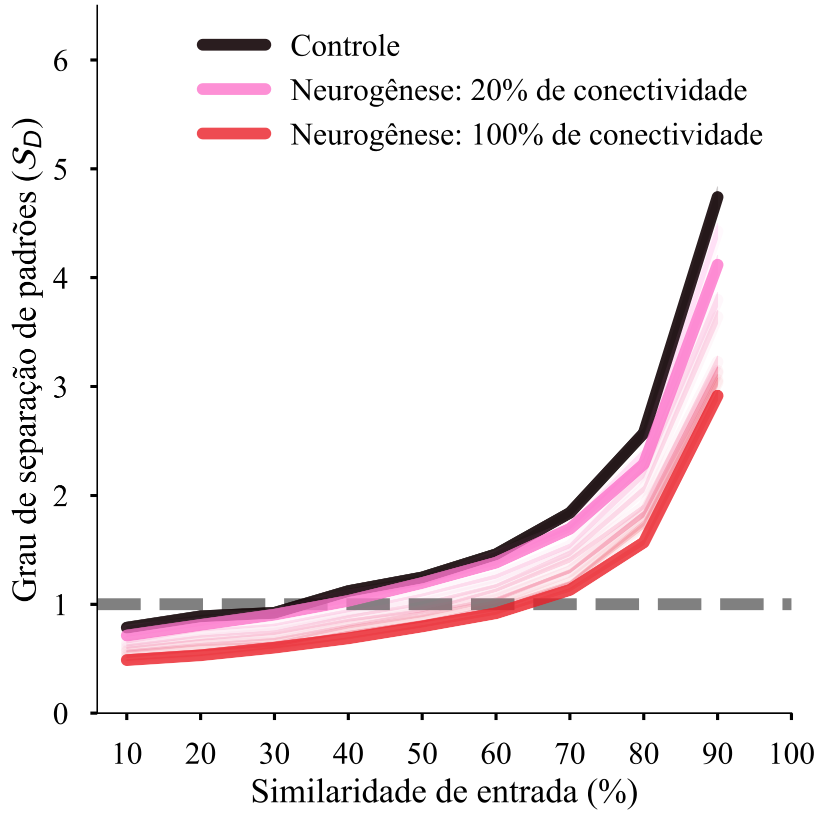


Figure 7 deepens the analysis by detailing the degree of pattern separation ($\mathcal{S}_D$) as a function of input pattern similarity. A value of $\mathcal{S}_D > 1$ means that the output representation in the DG is more distinct than the input representation in the EC. In general, all models, including the control, achieve higher pattern separation for highly similar inputs. This occurs because, mathematically, small differences in output encoding are amplified when input overlap is very large. The control model's performance with low-similarity inputs was suboptimal, failing to separate patterns effectively, which can again be attributed to insufficient network sparsity, as discussed earlier. The introduction of neurogenesis exacerbates this limitation: as iGC connectivity increases, the network's pattern separation capacity progressively decreases. The model with 100% iGC connectivity exhibits the worst performance under all similarity conditions. The underlying mechanism of this deterioration is the non-selective activation of iGCs, which, due to their high excitability, tend to form the same subset of active neurons in response to distinct input patterns, reducing the dissimilarity of output representations and, consequently, the degree of pattern separation.

To dissect the individual contributions of each neuronal population, Figure 8 presents the mean $\mathcal{S}_D$, aggregated across all similarity levels. This analysis confirms the previously observed trend, showing that the overall GC pattern separation capacity decreases with increasing iGC connectivity. Analysis of the subpopulations reveals a clear functional dichotomy: iGCs, in

Figura 7 – Degree of pattern separation ($\mathcal{S}_D$) per model and input similarity level. The dashed line represents a separation degree of 1; values above this indicate that pattern separation occurred, while values below indicate that it did not. Error is represented by the shaded area.

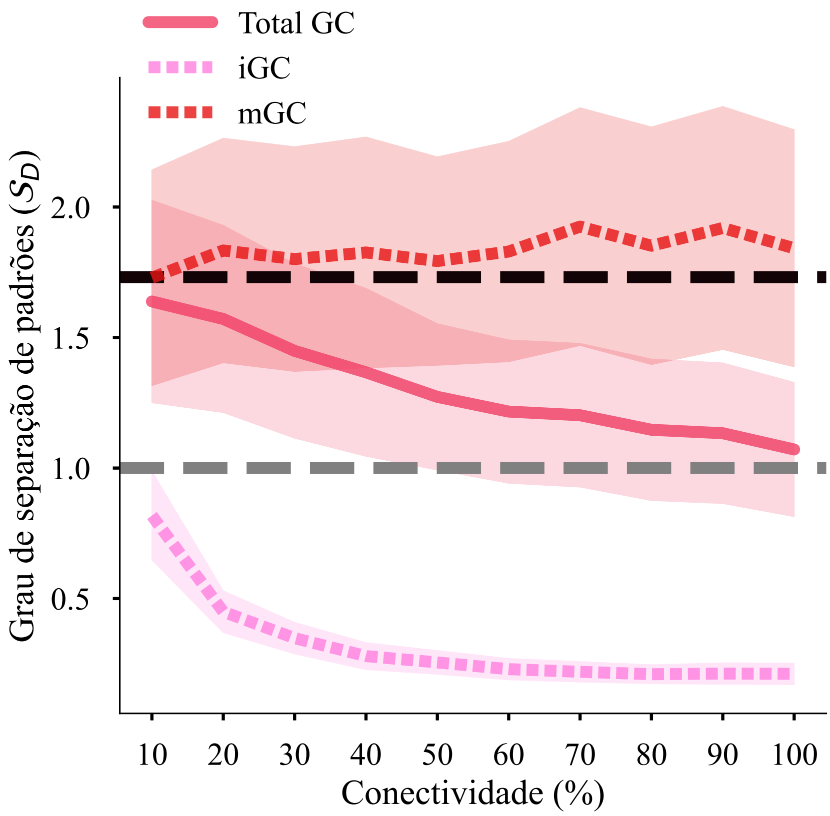


isolation, never achieve $\mathcal{S}_D > 1$, indicating that, rather than separating, they integrate input patterns, making them more similar. In contrast, mGCs demonstrate a robust pattern separation capacity across all models.

mGC performance improves slightly with increasing iGC connectivity and activity, likely a byproduct of increased global inhibition in the network, which, though insufficient, may be rendering mGC activity marginally more sparse. A simple linear regression model was used to evaluate the relationship between iGC connectivity level and mGC degree of pattern separation. The model found a significant, though weak, relationship between connectivity level and degree of pattern separation ($F(1, 8) = 7.83, p = 0.023, R^2 = 0.49, \mathcal{S}_D^{mGC} = 1.76 + 0.13 \times \text{Connectivity}$).

These findings corroborate the theory proposed by Kim e Lim (2024b), which suggests a role for mGCs as pattern separators and for iGCs as integrators. It remains to be investigated, in future experiments, what the impact of this mixed coding and the strengthening of separation in mGCs has on CA3 memory functions.

Figura 8 – Mean degree of pattern separation ($\mathcal{S}_D$) per model and population. The black dashed line represents the $\mathcal{S}_D$ of the control, while the gray one represents a separation degree of 1. Error is represented by the shaded area; in this case, the error is large given that $\mathcal{S}_D$ varies substantially across different similarity levels in Figure 7.



6.2 Expected results

It is expected that the presence of iGCs will impair pattern completion in CA3, due to increased activation of neural assemblies unrelated to the input pattern, an effect consistent with experimental results suggesting non-selective assembly activation by iGCs due to their high excitability (Ko et al., 2025). Additionally, in the temporal maturation simulation, iGCs are expected to encode patterns over time, integrating information presented while they were young, corroborating their role as temporal integrators, as proposed by (Aimone et al., 2009). It is therefore theorized that mGCs would play a predominant role in the separation of specific, distinct patterns, while iGCs, over the course of their maturation, would specialize in the separation of patterns close in time, specifically, those they encoded during their immature phase.

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